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2026 IEEE 5th World Conference on Applied Intelligence and Computing : Submission (2589) has been created.

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Track Name: AIC2026

Paper ID: 2589

Paper Title: IntelliDex: AI and Blockchain-Powered Trading Platform with Integrated Cybersecurity Framework

Abstract:
Trust remains one of the most persistent and overlooked problems in cryptocurrency price prediction. In many existing systems, prediction services are centralized, which creates several issues. There is a lack of data integrity, as prediction results can be selectively displayed or modified leading to misleading performance claims. The absence of transparency and verifiable audit trails makes it difficult for users to trust the authenticity of historical data. From a broader cyber security perspective, these platforms may also suffer from weak authentication and cyber control mechanisms, data tampering and account compromise. IntelliDex is our attempt to tackle this head-on — a Bitcoin price prediction system where every prediction is cryptographically locked in before the market moves, making manipulation practically impossible. At its core, the system uses a 4-layer Multi-Task Transformer with multi-head attention, drawing on 61 input features and 16 pulled from sentiment analysis to simultaneously predict price movement and direction across seven time horizons, ranging from 15 minutes all the way out to 7 days. Before any outcome is known, predictions are hashed using SHA-256 and written to Ethereum smart contracts, producing a tamper-proof, publicly accessible audit trail. Storage costs were a real concern. A hybrid on-chain/IPFS architecture brought those costs down by 90% compared to storing everything on-chain, without sacrificing cryptographic integrity. For transaction security, an Isolation Forest model handles anomaly detection, hitting 94.2% precision in our tests. On BTC/USDT, short-range directional accuracy clears 70% at the 15-minute mark, though — predictably — this drops to around 56% at the 7-day horizon. MAPE ranges from 1.5% to 7.0% depending on the timeframe. Each prediction commitment costs 42,183 gas units.

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